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Q1 B1

4 points

Q1.1 (a)

2 points

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(a) Show that $f'(x) = \frac{3e^{\frac{x}{3}}(x+2)}{x^{\frac{4}{3}}}$. Then find and classify all local extrema of $f(x)$.



1

+2.0



Correct. The correct answer is that $x = -2$ is a local max and $x = 0$ is a local minimum.



2

+0.5

The derivative is correctly computed and simplified.



3

+0.5

The correct critical points are identified ($x = 0$ and $x = -2$).

⋮ 4 +0.5

The correct sign chart is constructed.

⋮ 5 +0.5

The correct interpretation is made from the sign chart.

⋮ 6 +0.0

Little to no progress is made toward a correct answer.

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Q1.2 (b)

1 point

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(b) Find all points of inflection of $f(x)$. Be sure to fully justify your answer.

⋮ 1 +1.0

Correct. The correct answer is that both $-2 + \sqrt{6}$ and $-2 - \sqrt{6}$ are points of inflection.



2

+0.5

Some meaningful progress is made toward a solution, but there are still significant mistakes.



3

+0.0

Little to no progress is made toward a solution.

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Q1.3 (c)

1 point

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(c) Find the values of the absolute extrema of $f(x)$ on the domain $[-2, 2]$. You do not need to simplify your answers.



1

+1.0

Correct. The answer is $x = 0$ is the absolute minimum and $x = 2$ is the absolute maximum.



2

+0.5

Some meaningful progress is made toward a solution, but there are still significant mistakes.



3

+0.0

Little to no progress is made toward a correct answer.

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Q2 B2

4 points

Q2.1 (a)

1 point

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(a) Find all values c guaranteed by the Mean Value Theorem when applied to the function $f(x) = \frac{x^3}{3}$ on the interval $[-\sqrt{3}, \sqrt{3}]$.



1

+1.0

Correct. The answer shows that $c = \pm 1$ are the two values where the slope of the tangent line equals the slope of the secant line.



2

+0.5

The slope of the secant line is found to be 1.



3

+0.5

The equation $x^2 = m$ is correctly solved, where m is the result the student found for the slope of the secant line.



4

+0.0

Little to no progress is made toward a solution.

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Q2.2 (b)

3 points

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(b) Show that the function $f(x) = x^4 + e^x - x$ has exactly one point where the slope of the tangent line is 2. Be sure to fully justify your answer.



1

+3.0

Correct.



2

+0.5

The correct equation is written down to solve:

$$4x^3 + e^x - 3 = 0 \text{ or } 4x^2 + e^x - 1 = 2.$$



3

+0.5

Two values of x are shown to indicate the equation written down must have a solution. Credit should be awarded if this is done correctly even with a wrong equation.



4

+0.5

It is stated that since the $f'(x)$ is continuous, the IVT guarantees some solution within the interval.



5

+0.5

The correct second derivative is found: $12x^2 + e^x$.



6

+0.5

It is correctly reasoned that the second derivative has no roots (always positive).



7

+0.5

It is stated that since $12x^2 + e^x = 0$ has no solution, then $4x^3 + e^x - 3 = 0$ has at most one solution (by MVT/Rolle/Rolle's corollary, which applies because $f'(x)$ is differentiable).



8

+0.0

Little to no progress is made toward a solution.

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Q3 B3

4 points

Q3.1 (a)

1 point

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(a) Find all horizontal asymptotes of $f(x)$.

 1 **+1.0**

Correct. The horizontal asymptotes are $y = 1$ as $x \rightarrow \infty$ and $y = -1$ as $x \rightarrow -\infty$.

 2 **+0.5**

Some meaningful progress is made toward a solution, but there are still significant errors.

 3 **+0.0**

No meaningful progress is made.

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Q3.2 (b)

1 point

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(b) Compute two one-sided limits at each vertical asymptote of $f(x)$.



1

+1.0

Correct. Both one-sided limits at $x = -1$ tend to $-\infty$.



2

+0.5

Some meaningful progress is made toward a solution, but there are still significant errors.



3

+0.0

No meaningful progress is made.

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Q3.3 (c)

1 point

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(c) Find all intervals where $f(x)$ is decreasing.



1

+1.0

Correct. $f(x)$ is decreasing on $(-\infty, -1)$.



2

+0.5

Some meaningful progress is made toward a solution, but there are still significant errors.

3 +0.0

No meaningful progress is made.

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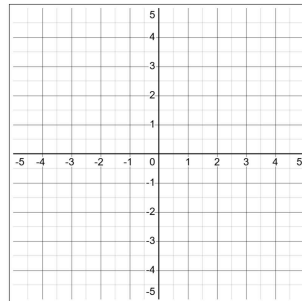
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Q3.4 (d)

1 point

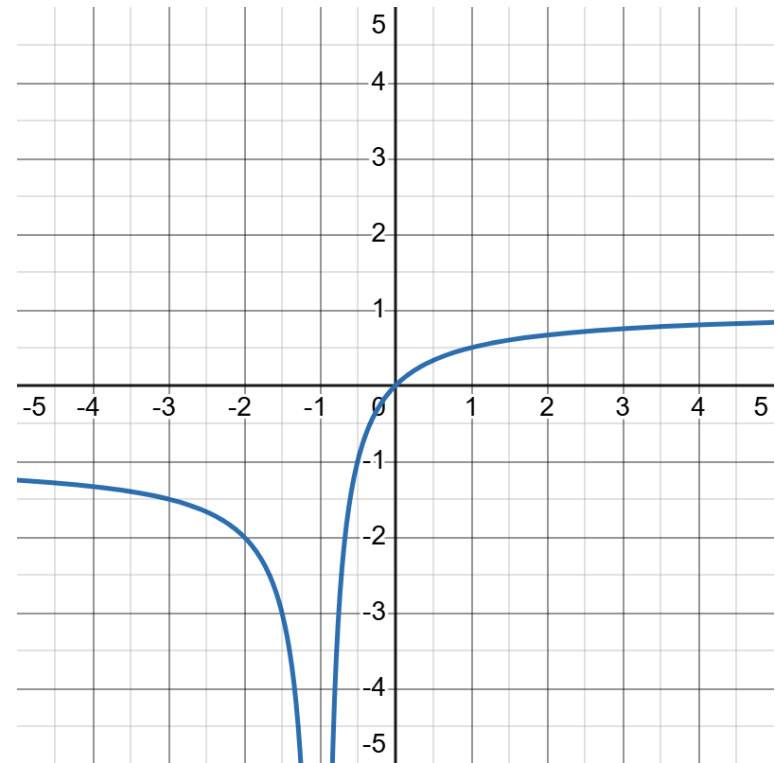
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- (d) Use the information from the previous parts of the problem to sketch an accurate graph of $f(x)$ on the axes shown to the right. Be sure that your sketch accurately reflects any intercepts.



1 +1.0

Correct. The graph is shown below. Credit should be awarded if this graph follows correctly from an incorrect previous part.



2 +0.5

Some meaningful progress is made toward a solution, but there are still significant errors.

3 +0.0

No meaningful progress is made.

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Q4 B4

4 points

Q4.1 (a)

1.5 points

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(a) Consider the functions $f(x) = x^3$ and $g(x) = 27x$. What is the smallest vertical distance between these functions on the interval $[1, 5]$? Be sure to fully justify your answer.

⋮ 1 +1.5

Correct. The smallest vertical distance is 10 which occurs at $x = 5$.

⋮ 2 +0.5

The correct distance function is found:

$$D(x) = 27x - x^3.$$

⋮ 3 +0.5

The critical point is found: $x = 3$. Credit is awarded if a critical point is found correctly from the wrong distance function.

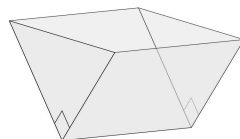
⋮ 4 +0.5

The critical point and endpoints are tested to find the absolute minimum. Credit is awarded if this part follows correctly from an incorrect previous part.

Q4.2 (b)

2.5 points

- (b) A trough with no top is built with two isosceles right triangles as ends, as shown on the right. If the volume of the trough must be 4 cubic meters, find the smallest area of material required to build it. The formula for the volume of the trough is the area of one of the triangles times the trough's length. Be sure to fully justify your answer and include units.



5 +0.0

No meaningful progress is made toward an answer.

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1 +2.5

Correct. The answer is 12 square meters.

2 +0.5

The correct expression for surface area is used: $2lx + x^2$.

3 +0.5

The relationship $4 = \frac{1}{2}x^2l$ is used to substitute into the surface area formula.

4 +0.5

The correct critical point is found from the surface area function: $x = 2$.

⋮ 5 +0.5

It is correctly reasoned that the critical point is an absolute minimum, as it is a local minimum and the only critical point on the interval.

⋮ 6 +0.5

The correct numerical value is reported along with the correct units (square meters).

⋮ 7 +0.0

No meaningful progress is made toward an answer.

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Q5 B5

4 points

Q5.1 (a)

2 points

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(a) Suppose that $f(x)$ and $g(x)$ are two functions defined on an interval such that $f'(x) = g'(x)$. Suppose also that $f''(x)$ exists, and that $F(x)$ is an antiderivative of $f(x)$ and $G(x)$ is an antiderivative of $g(x)$. Which of the following must be true? Circle all that apply.

- A. $f(a) - g(a) = f(b) - g(b)$ for any a and b in the interval.
- B. $f(b) - f(a) = g(b) - g(a)$ for any a and b in the interval.
- C. $f(a) = g(a)$ for any a in the interval.
- D. $f''(a) = g''(a)$ for any a in the interval.
- E. $F''(a) = G''(a)$ for any a in the interval.
- F. $F(x) = G(x) + C$ for some constant C .
- G. The graph of $f(x)$ is a vertical stretch of the graph of $g(x)$.
- H. $a \cdot F(x) - G(x) + b$ is an antiderivative of $a \cdot f(x) - g(x) + b$ for any constants a and b .



1

+2.0

Correct. Only A, B, D, and E are true.



2

+1.5

7 out of 8



3

+1.0

6 out of 8



4

+0.5

5 out of 8



5

+0.0

4 or fewer out of 8

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Q5.2 (b)

2 points

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(b) Suppose $h(x)$ is a function such that $h(1) = 2$, $h(-2) = -1$ and $h''(x) = 6$. Find two different antiderivatives of $h(x)$.



1

+2.0

Correct. An antiderivative of $h(x)$ is anything of the form $x^3 + 2x^2 - 5x + C$.



2

+0.5

The solution correctly finds $h(x) = 3x^2 + Ax + B$ for some constants A and B .



3

+0.5

The solution correctly uses the initial values to setup a system of equations for A and B . Credit is awarded if this follows correctly from an incorrect previous part.



4

+0.5

The system of equations is solved correctly (even if it is the wrong system of equations).



5

+0.5

After finding A and B , the solution takes another antiderivative and chooses any two values for C .



6

+0.0

No meaningful progress is made toward an answer.

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